Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C

Chapter 5
Memory Access

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Acknowledgement: Lecture slides based on Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C, University of Maine https://web.eece.maine.edu/~zhu/book/

Overview

- How data is organized in memory?
  - Big-Endian vs Little-Endian

- How data is addressed?
  - Register offset
    - `LDR r1, [r0, r3]` ; offset = r3
    - `LDR r1, [r0, r3, LSL #2]` ; offset = r3 * 4
  - Immediate offset
    - Pre-index: `LDR r1, [r0, #4]`
    - Post-index: `LDR r1, [r0], #4`
    - Pre-index with update: `LDR r1, [r0, #4]!`

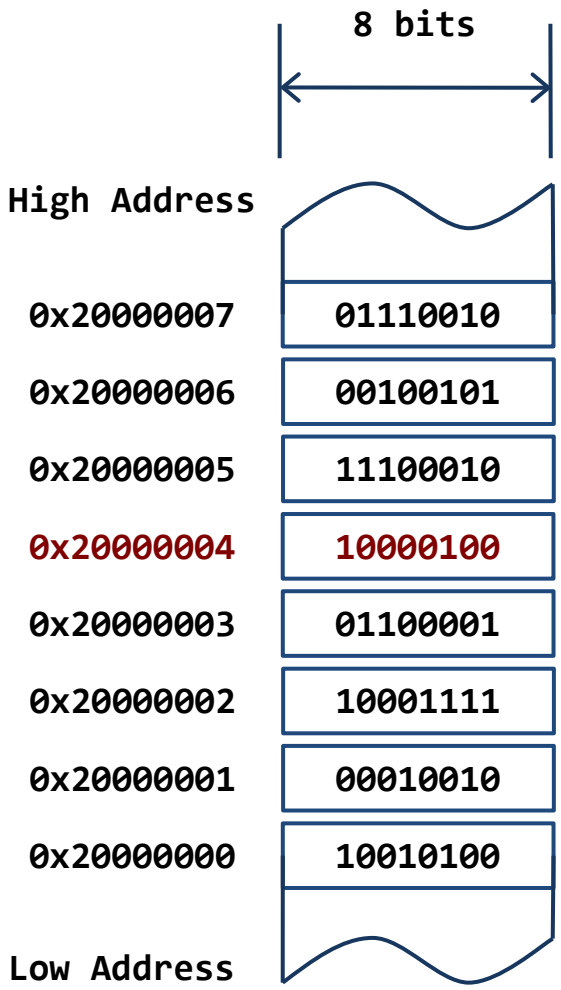
Logic View of Memory

- By grouping bits together we can store more values
  - 8 bits = **1 byte**
  - 16 bits = 2 bytes = **1 halfword**
  - 32 bits = 4 bytes = **1 word**
- From software perspective, memory is an addressable array of bytes.
  - The byte stored at the memory address `0x20000004 (hex, 4 bytes)` is `0b10000100 (binary, 1 byte)`

`0b10000100` → `0x84` → `132`

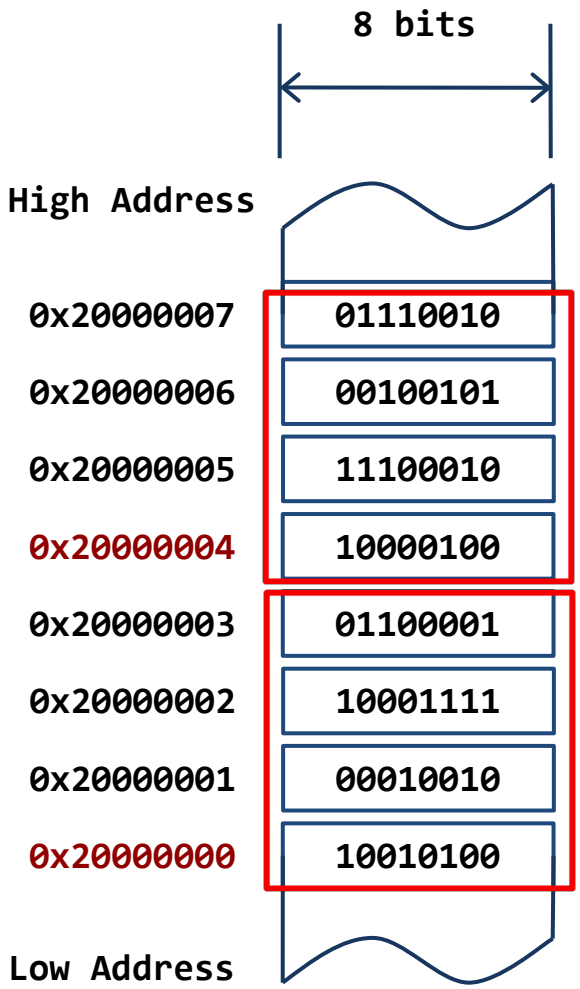
Binary Hexadecimal Decimal

Computer memory is *byte-addressable*!



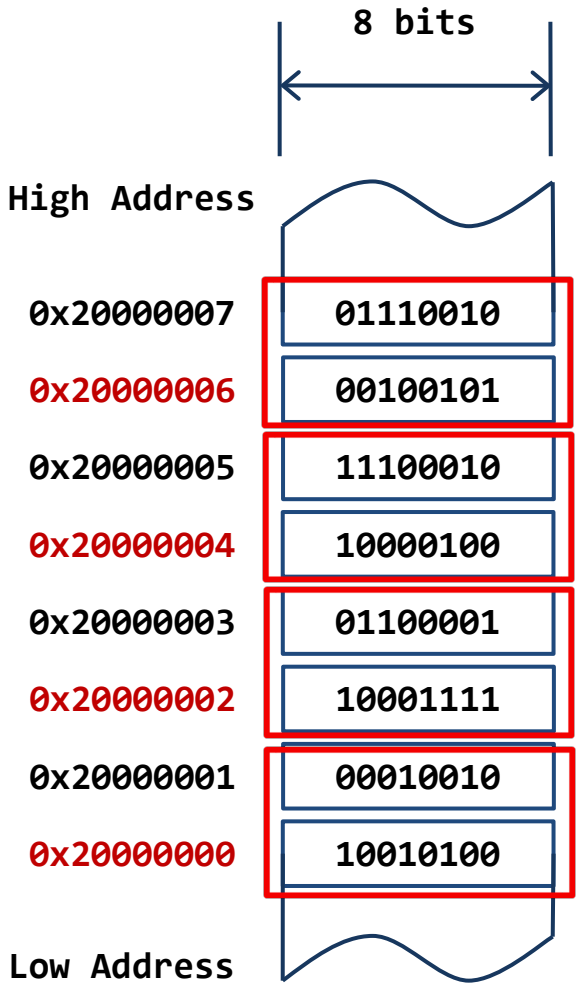
Logic View of Memory

- When we refer to memory locations by address, we can only do so in units of bytes, halfwords or words
- Architecturally aligned addresses are required for efficient access; unaligned access may require multiple cycles
  - A word should be stored at an address that's divisible by 4 (Word-address mod 4 = 0, binary address ends with 00)
    - Memory address of a word is the lowest address of all four bytes in that word.
    - Two words at addresses: `0x20000000` and `0x20000004`
  - A halfword should be stored at an address that's divisible by 2 (Halfword-address mod 2 = 0, binary address ends with 0)
    - Memory address of a halfword is the lowest address of all 2 bytes in that halfword.



Logic View of Memory

- Halfwords
  - **16** bits = **2** bytes = **1** halfword
  - The right diagram has four halfwords at addresses of:
    - 0x20000000
    - 0x20000002
    - 0x20000004
    - 0x20000006



Data Alignment and Performance

- Assume a byte-addressable memory with a data bus that is 32 bits (4 bytes) wide
- Consider 16 bytes of memory (addresses 0 to 15) arranged as four 32-bit words (4 bytes each)

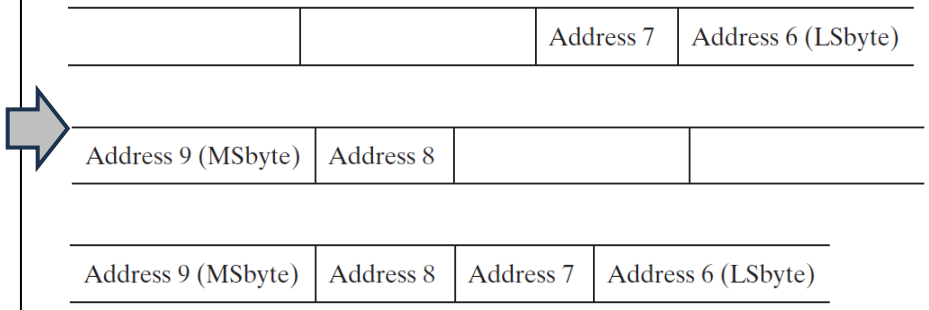
Address 15	Address 14	Address 13	Address 12
Address 11	Address 10	Address 9	Address 8
Address 7 (MSbyte)	Address 6	Address 5	Address 4 (LSbyte)
Address 3	Address 2	Address 1	Address 0

Well-aligned: each word begins on a mod-4 address, which can be read in a single memory cycle

Address 15	Address 14	Address 13	Address 12
Address 11	Address 10	Address 9 (MSbyte)	Address 8
Address 7	Address 6 (LSbyte)	Address 5	Address 4
Address 3	Address 2	Address 1	Address 0

Ill-aligned: a word begins on address 6, not a mod-4 address, which can be read in 2 memory cycles

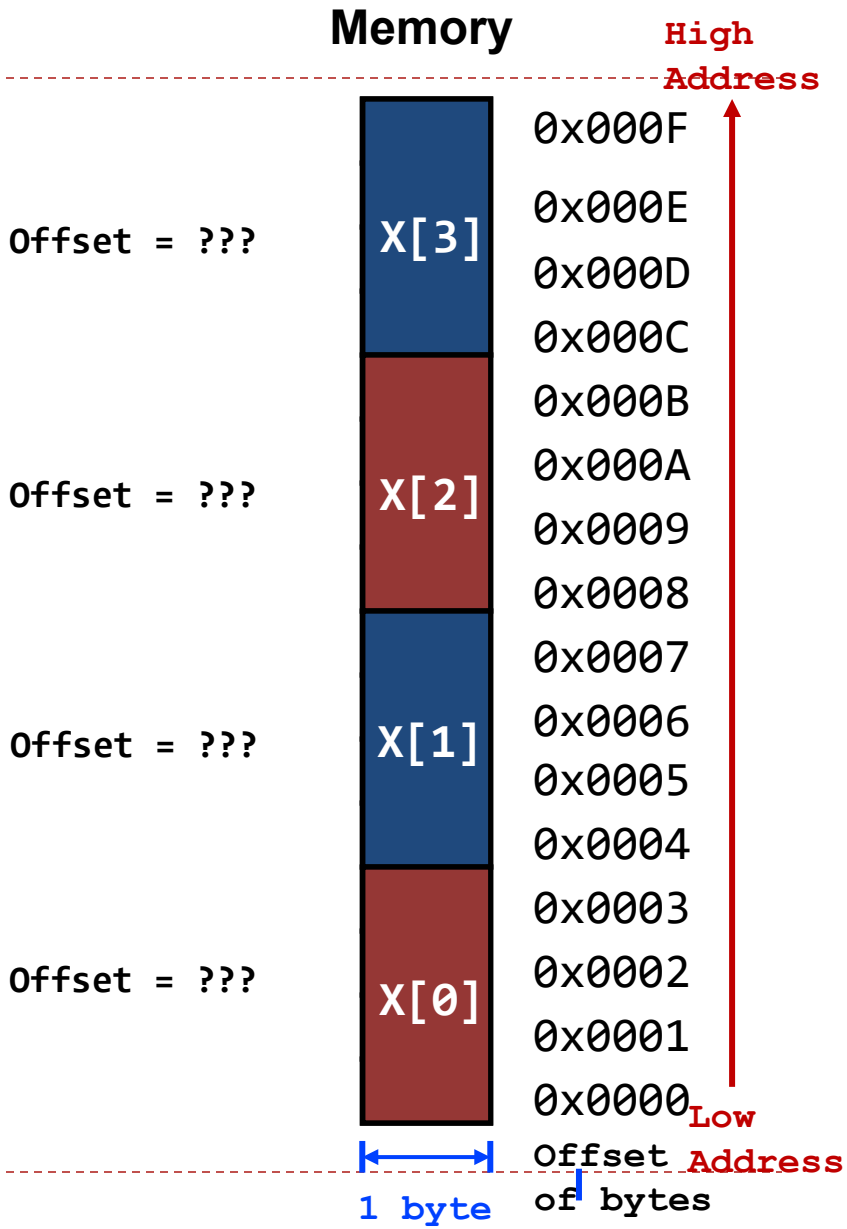
The first read cycle would retrieve 4 bytes from addresses 4 through 7; of these, the bytes from addresses 4 and 5 are discarded, and those from addresses 6 and 7 are moved to the far right;
The second read cycle retrieves 4 bytes from addresses 8 through 11; the bytes from addresses 10 and 11 are discarded, and those from addresses 8 and 9 are moved to the far left;
Finally, the two halves are combined to form the desired 32-bit operand.



Quiz

```
uint32_t X[4];
```

What are their memory address offsets?



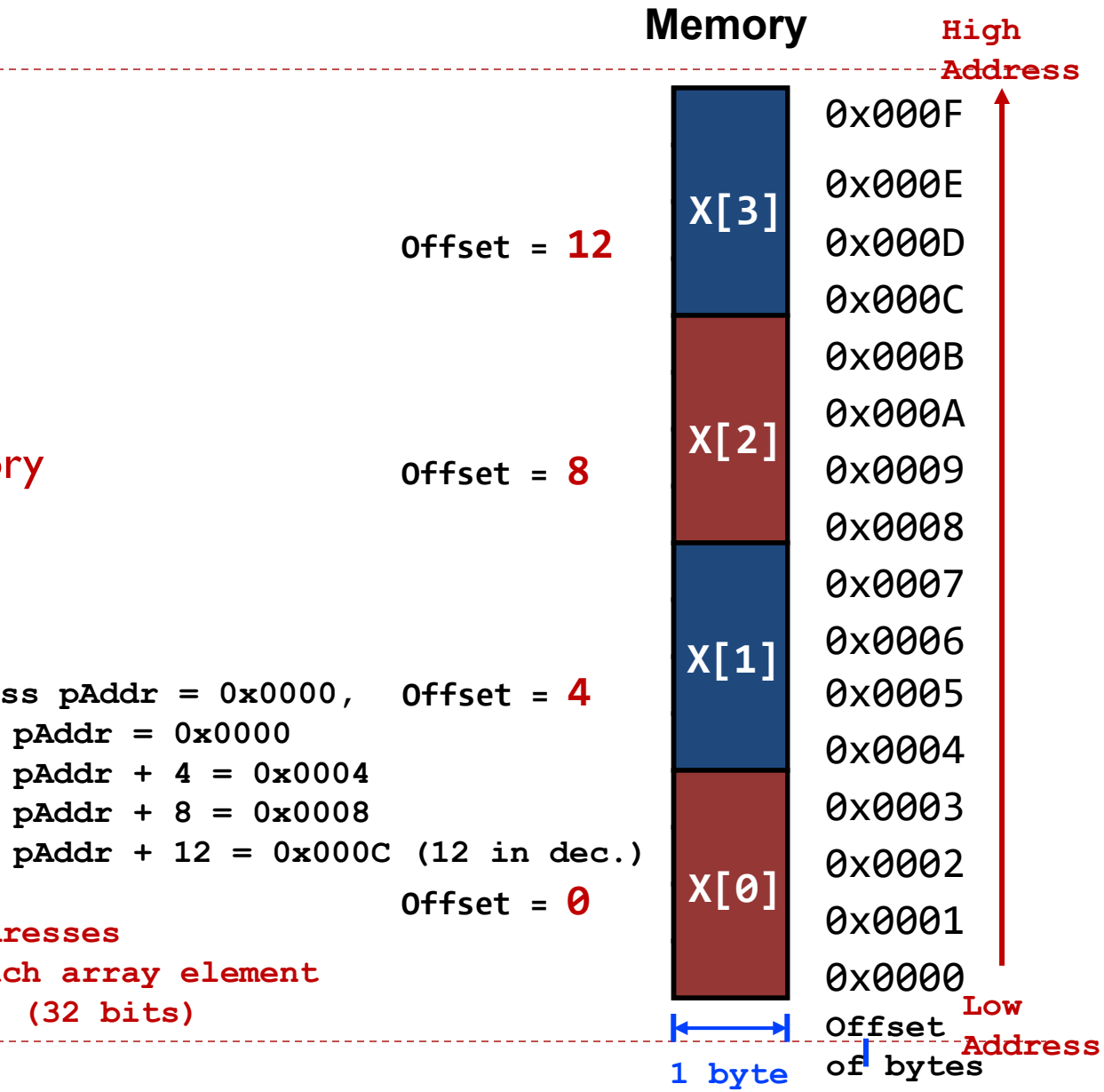
Quiz ANS

```
uint32_t X[4];
```

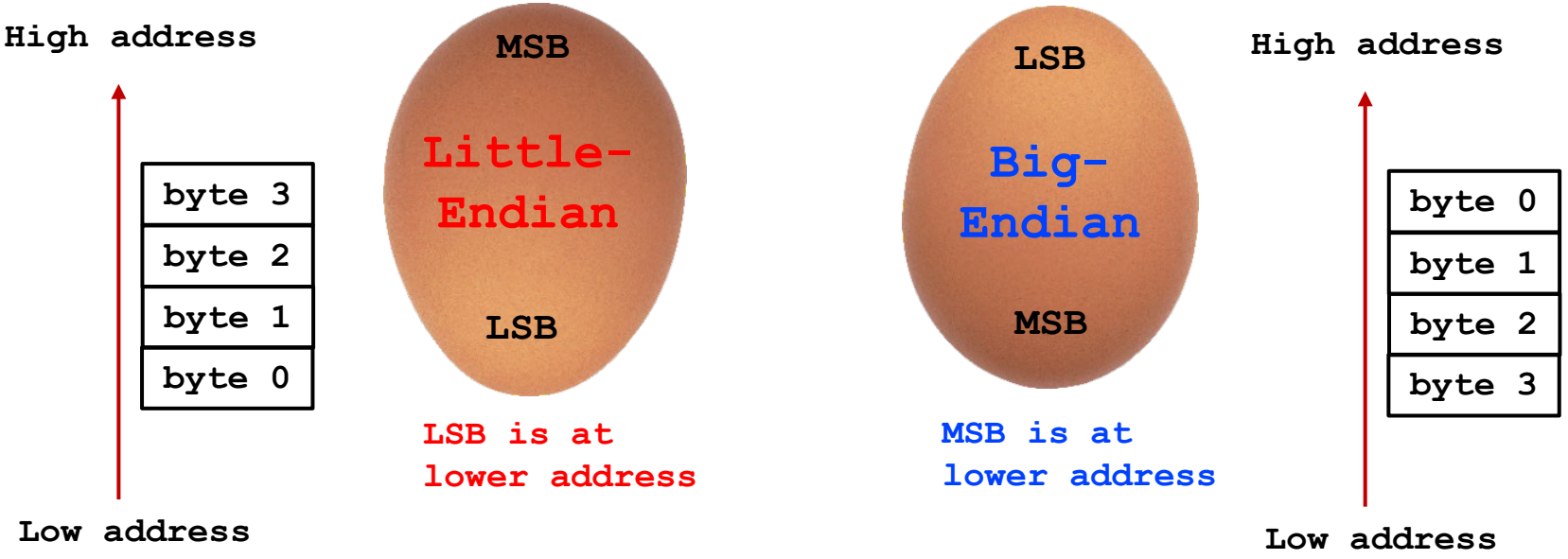
What are their memory address offsets?

If the array starts at address pAddr = 0x0000,
- Memory address of X[0] is pAddr = 0x0000
- Memory address of X[1] is pAddr + 4 = 0x0004
- Memory address of X[2] is pAddr + 8 = 0x0008
- Memory address of X[3] is pAddr + 12 = 0x000C (12 in dec.)

Sequential words are at addresses incrementing by 4, since each array element of type uint32_t is 4 bytes (32 bits)



Endianess



Gulliver's Travels (by Jonathan Swift, published in 1726):

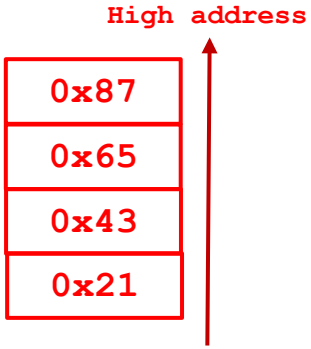
- Two religious sects of Lilliputians
- The Little-Endians crack open their eggs from the little end
- The Big-Endians break their eggs on the big end

Endianess

Endian: byte order, not bit order!



Little-Endian




```
uint32_t a = 0x87654321
```

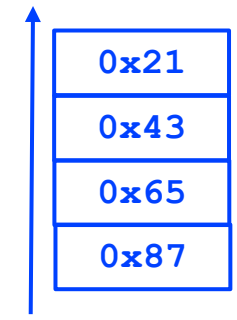
Reading from the top

byte 3	byte 2	byte 1	byte 0
0x87	0x65	0x43	0x21

byte 0	byte 1	byte 2	byte 3

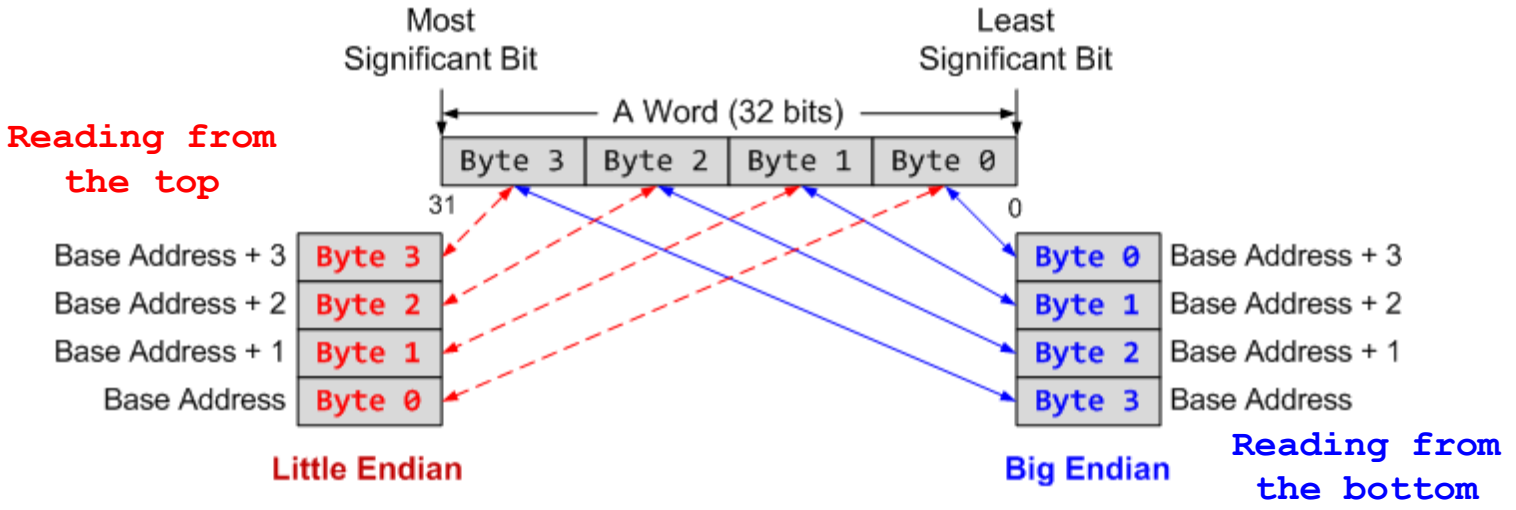
Reading from the bottom

Big-Endian



Endianess

- Little-Endian
  - Least significant byte (LSB) is stored at lowest (least) address of a word
- Big-Endian
  - Most significant byte (MSB) is stored at lowest (least) address of a word
- Regardless of endianness, the address of a word is defined as the lowest address of all bytes it occupies.
- ARM is *Little-Endian by default*.
  - It can be made Big-Endian by configuration.



Endianness Example

- **Little-Endian**
  - **LSB is at lower address**

```
                                Memory     Value
                                Offset   (LSB) (MSB)
                                ======   ===========
uint8_t a  = 1;                 0x0000   01 02 FF 00
uint8_t b  = 2;
uint16_t c = 255; // 0x00FF
uint32_t d = 0x12345678;        0x0004   78 56 34 12
                                      Increasing mem. addr.
```

- **Big-Endian**
  - **MSB is at lower address**

```
                                Memory     Value
                                Offset   (MSB) (LSB)
                                ======   ===========
uint8_t a  = 1;                 0x0000   01 02 00 FF
uint8_t b  = 2;
uint16_t c = 255; // 0x00FF
uint32_t d = 0x12345678;        0x0004   12 34 56 78
                                      Increasing mem. addr.
```

- **For uint8_t a and b, each with size of 1 Byte: No difference**
- **Little-endian:**
  - **For uint16_t c with size of 2 Bytes: LSB FF is at lower address and MSB 00 is at higher address**
  - **For uint32_t d with size of 4 Bytes: LSB 78 is at lower address and MSB 12 is at higher address.**
- **Big-endian:**
  - **For uint16_t c with size of 2 Bytes: LSB FF is at higher address and MSB 00 is at lower address**
  - **For uint32_t d with size of 4 Bytes: LSB 78 is at higher address and MSB 12 is at lower address.**

Example

If Big-Endian is used, the word stored at address 0x20008000 is

If Little-Endian is used, the word stored at address 0x20008000 is

Memory Address	Memory Data
0x20008003	0xA7
0x20008002	0x90
0x20008001	0x8C
0x20008000	0xEE

Example

If Big-Endian is used, the word stored at address 0x20008000 is

0xEE8C90A7

If Little-Endian is used, the word stored at address 0x20008000 is

0xA7908CEE

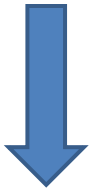
Endianness specifies byte order, not bit order in a byte!

Memory Address	Memory Data
0x20008003	0xA7
0x20008002	0x90
0x20008001	0x8C
0x20008000	0xEE

Load-Modify-Store

C statement

```
x = x + 1;
```

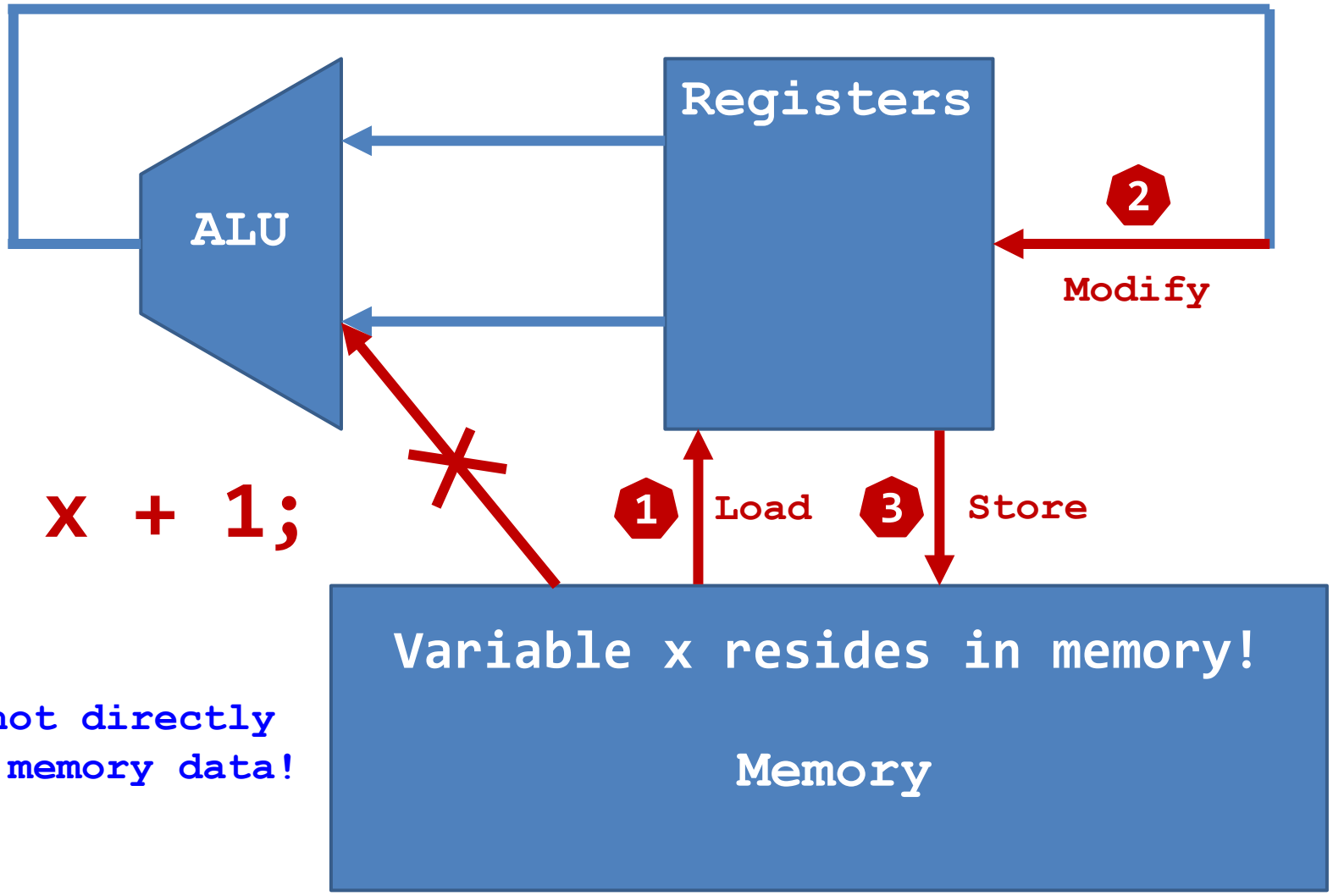


Assume variable X resides in memory and is a 32-bit integer

```
; Assume the memory address of x is stored in r1

LDR r0, [r1]      ; load value of x from memory
ADD r0, r0, #1    ; x = x + 1
STR r0, [r1]      ; store x into memory
```

3 Steps: Load, Modify, Store



ALU cannot directly operate memory data!

Load Instructions

- **LDR rt, [rs]**
  - **Read from memory**
  - Mnemonic: **L**oa**D** to **R**egister (**LDR**)
  - rs specifies the memory address
  - rt holds the 32-bit value fetched from memory

  - For Example:

```
; Assume r0 = 0x08200004
; Load a word:
LDR r1, [r0]          ; r1 = Memory.word[0x08200004]
```

Store Instructions

- **STR rt, [rs]**
  - **Write into memory**
  - Mnemonic: **ST**ore from **R**egister (**STR**)
  - rs specifies memory address
  - Save the content of rt into memory

  - For Example:

```
; Assume r0 = 0x08200004
; Store a word
STR r1, [r0]     ; Memory.word[0x08200004] = r1
```

Load/Store a Byte, Halfword, Word

```
LDRxxx R0, [R1]
; Load data from memory into a 32-bit register
```

LDR	Load Word	uint32_t/int32_t	unsigned or signed int
LDRB	Load **B**yte	uint8_t	unsigned char
LDRH	Load **H**alfword	uint16_t	unsigned short int
LDRSB	Load **S**igned **B**yte	int8_t	signed char
LDRSH	Load **S**igned **H**alfword	int16_t	signed short int

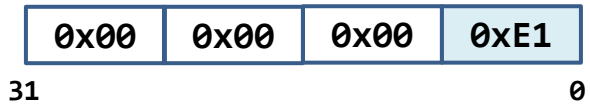
```
STRxxx R0, [R1]
; Store data extracted from a 32-bit register into memory
```

STR	Store Word	uint32_t/int32_t	unsigned or signed int
STRB	Store Lower **B**yte	uint8_t/int8_t	unsigned or signed char
STRH	Store Lower **H**alfword	uint16_t/int16_t	unsigned or signed short

Load a Byte, Half-word, Word (Little-Endian)

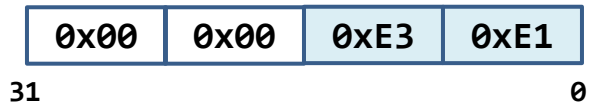
Load a Byte

```
LDRB r1, [r0]
```



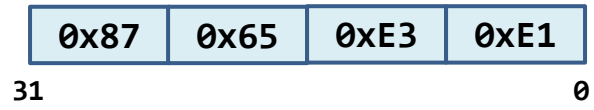
Load a Halfword

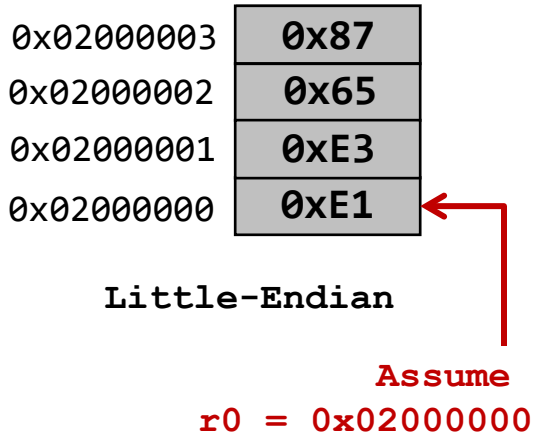
```
LDRH r1, [r0]
```



Load a Word

```
LDR r1, [r0]
```





LDRH "Load Register Halfword": it loads a 16-bit halfword value from the memory address pointed to by register r0 into register r1. The loaded 16-bit value is zero-extended to fill the 32-bit register r1. This means the upper 16 bits of r1 will be set to zero regardless of the halfword data.
LDRB "Load Register Byte": it loads 8-bit byte value from the memory address pointed to by register r0 into register r1, and zero-extends it.

Sign Extension (Little-Endian)

Load a Signed Byte

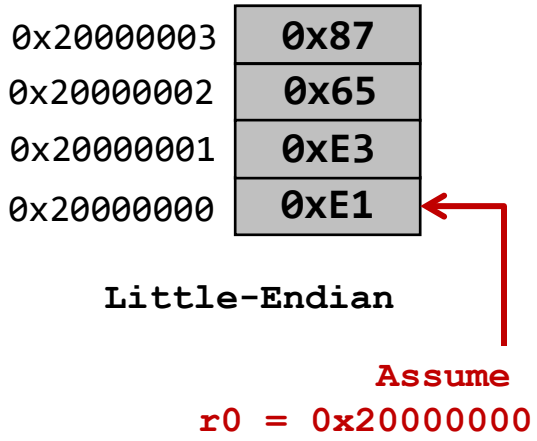
```
LDRSB r1, [r0]
```



Load a Signed Halfword

```
LDRSH r1, [r0]
```





LDRSH "Load Register Signed Halfword"
LDRSB "Load Register Signed Byte"
Similar to LDRH and LDRB, except each sign-extends the value to fill the 32-bit register, not zero-extend. Facilitate subsequent 32-bit signed arithmetic.

Address Modes: Offset in Register

- Address accessed by **LDR/STR** is specified by a base register plus an offset
- Offset can be held in **a register**

```
LDR r0,[r1,r2]
```

- Base memory address held in register r1
- Offset held in r2
- Target address = r1 + r2

```
LDR r0,[r1,r2,LSL #2]
```

- Base memory address held in register r1
- Offset = r2, LSL #2
- Target address = r1 + r2 * 4

Address Modes: Immediate Offset

- Address accessed by **LDR/STR** is specified by a base register plus an offset
- Offset can be **an immediate value**

```
LDR r0,[r1,#8]
```

- Base memory address held in register r1
- Offset is an immediate value
- Target address = r1 + 8

Three modes for immediate offset:

- Pre-index,
- Post-index,
- Pre-index with Update

Addressing Mode: Pre-index *vs* Post-index

- Pre-index

```
LDR r1, [r0, Offset]
```

- Post-index

```
LDR r1, [r0], Offset
```

- Pre-index with Update

```
LDR r1, [r0, Offset]!
```

The table assumes r0 = 0x100, offset = 4 bytes (#4)

Mode	Address used for Load	Base register update	Example (r0=0x100)
Pre-index LDR r1, [r0, #4]	r0 + offset (0x104)	No	r1 = data[0x104]; r0 = 0x100
Post-index LDR r1, [r0], #4	r0 (0x100)	Yes, after load	r1 = data[0x100]; r0 = 0x104
Pre-index w/ Update LDR r1, [r0, #4]!	r0 + offset (0x104)	Yes, before load	r1 = data[0x104]; r0 = 0x104

Pre-index

Pre-Index: `LDR r1, [r0, #4]`

Offset: range is -255 to +255

Assume: r0 = 0x20008000

Memory Address	Memory Data
0x20008007	0x88
0x20008006	0x79
0x20008005	0x6A
0x20008004	0x5B
0x20008003	0x4C
0x20008002	0x3D
0x20008001	0x2E
0x20008000	0x1F

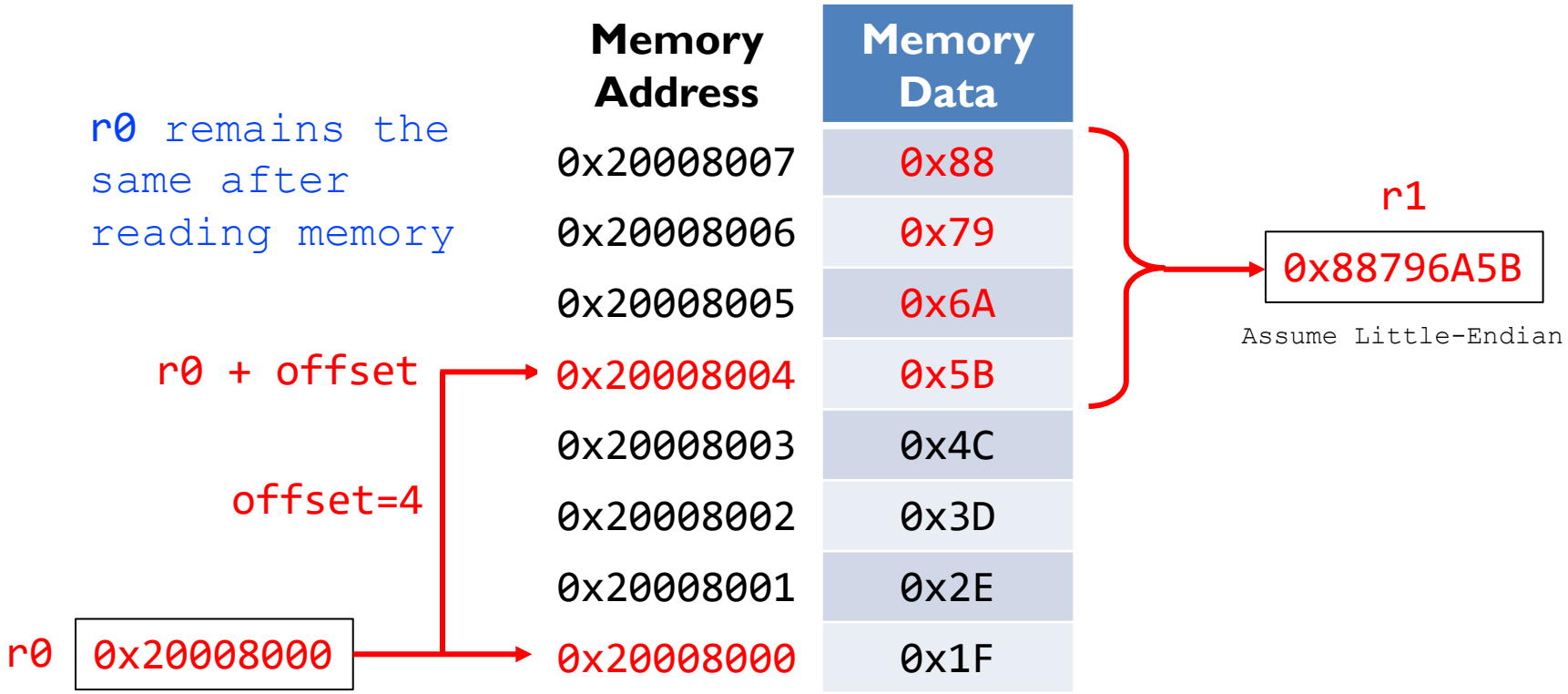
- Calculates address by adding the offset (here, #4) to the base register (r0) before the load. Loads data from the resulting address r0+4 into r1. The base register (r0) is not updated.
- Example: instruction accesses memory at r0 + 4 = 0x20008004, but r0 remains to be 0x20008000 after execution.

Pre-index

Pre-Index: LDR r1, [r0, #4]

Offset: range is -255 to +255

Assume: r0 = 0x20008000



Accessing an Array

- C code

```
uint32_t array[10];
array[0] += 5;
array[1] += 5;
```

Assume the memory address of the array starts at 0x20008000.

- Pre-index

Assume r0 = 0x20008000.

```
LDR r1, [r0]      ; Read array[0]
ADD r1, r1, #5
STR r1, [r0]      ; Write to array[0]

LDR r1, [r0, #4] ; Read array[1]
ADD r1, r1, #5
STR r1, [r0, #4] ; Write to array[1]
```

Post-index

Post-Index: `LDR r1, [r0], #4`

Offset: range is -255 to +255

Assume: r0 = 0x20008000

Memory Address	Memory Data
0x20008007	0x88
0x20008006	0x79
0x20008005	0x6A
0x20008004	0x5B
0x20008003	0x4C
0x20008002	0x3D
0x20008001	0x2E
0x20008000	0x1F

- Loads data from the address currently in r0 into r1. After the load, updates the base register (r0) by adding the offset (#4).
- Example: instruction accesses memory at r0 = 0x20008000, then increments r0 by the offset of 4 to r0 + 4 = 0x20008004 after execution.

Post-index

Post-Index: LDR r1, [r0], #4

Offset: range is -255 to +255

Assume: r0 = 0x20008000

Increment r0 by offset after reading memory

r0 = r0 + offset

Memory Address	Memory Data
0x20008007	0x88
0x20008006	0x79
0x20008005	0x6A
0x20008004	0x5B
0x20008003	0x4C
0x20008002	0x3D
0x20008001	0x2E
0x20008000	0x1F

r0 0x20008004

r1 0x4C3D2E1F

Assume Little-Endian

Pre-index with Update

Pre-Index with Update: `LDR r1, [r0, #4]!`

Offset: range is -255 to +255

Assume: r0 = 0x20008000

Memory Address	Memory Data
0x20008007	0x88
0x20008006	0x79
0x20008005	0x6A
0x20008004	0x5B
0x20008003	0x4C
0x20008002	0x3D
0x20008001	0x2E
0x20008000	0x1F

- First, adds the offset (#4) to the base register (r0), then loads from this updated address r0 + 4. Base register r0 is set to r0 + 4 afterwards.
- Example: instruction accesses memory at r0 + 4 = 0x20008004, and also sets r0 to 0x20008004 after execution.

Pre-index with Update

Pre-Index with Update: LDR r1, [r0, #4]!

Assume: r0 = 0x20008000

Offset: range is -255 to +255

Increment r0 by offset before reading memory

r0 = r0 + offset

r0 0x20008004

Memory Address	Memory Data
0x20008007	0x88
0x20008006	0x79
0x20008005	0x6A
0x20008004	0x5B
0x20008003	0x4C
0x20008002	0x3D
0x20008001	0x2E
0x20008000	0x1F

r1

0x88796A5B

Assume Little-Endian

Summary of Pre-index and Post-index

Index Format	Example	Equivalent
Pre-index	`LDR r1, [r0, #4]`	`r1 ← memory[r0 + 4],` `r0 is unchanged`
Post-index	`LDR r1, [r0], #4`	`r1 ← memory[r0]` `r0 ← r0 + 4`
Pre-index with update	`LDR r1, [r0, #4]!`	`r1 ← memory[r0 + 4]` `r0 ← r0 + 4`

- **Pre-index:** use (base + offset), don't change base
  Post-index: use base, then change base
  Pre-index with update: change base, then use it
- For the pre-/post-indexed (writeback) forms in the Cortex-M/Thumb-2 instruction set (and for halfword and signed byte/halfword load/store in A32), the offset is an unsigned 8-bit immediate 0–255, and the U bit selects addition or subtraction, yielding an effective signed range of [−255, +255] around the base register.
- In ARM (A32) instruction set, for word and unsigned byte LDR/STR, the immediate has an effective signed range of [−4095, +4095]

Example (Little-Endian ordering)

```
LDRH r1, [r0]
; r0 = 0x20008000
```

r1 before load

0x12345678

r1 after load

Memory Address	Memory Data
0x20008003	0x89
0x20008002	0xAB
0x20008001	0xCD
0x20008000	0xEF

Example ANS (Little-Endian ordering)

```
LDRH r1, [r0]
; r0 = 0x20008000
```

r1 before load (r1 is Overwritten by LDR)

0x12345678

r1 after load

0x0000CDEF

Memory Address	Memory Data
0x20008003	0x89
0x20008002	0xAB
0x20008001	0xCD
0x20008000	0xEF

Example (Endianness does not matter for single byte)

```
LDRSB r1, [r0]
; r0 = 0x20008000
```

r1 before load

0x12345678

r1 after load

Memory Address	Memory Data
0x20008003	0x89
0x20008002	0xAB
0x20008001	0xCD
0x20008000	0xEF

Example ANS (Endianness does not matter for single byte)

```
LDRSB r1, [r0]
; r0 = 0x20008000
```

r1 before load (r1 is Overwritten by LDR)

0x12345678

r1 after load

0xFFFFFFEF

Memory Address	Memory Data
0x20008003	0x89
0x20008002	0xAB
0x20008001	0xCD
0x20008000	0xEF

Example (Little-Endian ordering)

```
STR r1, [r0, #4]
; r0 = 0x20008000,  r1=0x76543210
```

r0 before the store

0x20008000

r0 after the store

	Memory Address	Memory Data
	0x20008007	0x00
	0x20008006	0x00
	0x20008005	0x00
	0x20008004	0x00
	0x20008003	0x00
	0x20008002	0x00
	0x20008001	0x00
r0 ⇨	0x20008000	0x00

Example ANS (Little-Endian ordering)

```
STR r1, [r0, #4]
; r0 = 0x20008000,  r1=0x76543210
```

r0 before store

0x20008000

r0 after store

0x20008000

	Memory Address	Memory Data
	0x20008007	0x76
	0x20008006	0x54
	0x20008005	0x32
	0x20008004	0x10
	0x20008003	0x00
	0x20008002	0x00
	0x20008001	0x00
r0 ⇨	0x20008000	0x00

Example (Little-Endian ordering)

```
STR r1, [r0], #4
; r0 = 0x20008000,  r1=0x76543210
```

r0 before store

0x20008000

r0 after store

	Memory Address	Memory Data
	0x20008007	0x00
	0x20008006	0x00
	0x20008005	0x00
	0x20008004	0x00
	0x20008003	0x00
	0x20008002	0x00
	0x20008001	0x00
r0 ⇨	0x20008000	0x00

Example ANS (Little-Endian ordering)

```
STR r1, [r0], #4
; r0 = 0x20008000,  r1=0x76543210
```

r0 before store

0x20008000

r0 after store

0x20008004

Memory Address	Memory Data
0x20008007	0x00
0x20008006	0x00
0x20008005	0x00
r0 ⇨ 0x20008004	0x00
0x20008003	0x76
0x20008002	0x54
0x20008001	0x32
0x20008000	0x10

Example (Little-Endian ordering)

```
STR r1, [r0, #4]!
; r0 = 0x20008000,  r1=0x76543210
```

r0 before store

0x20008000

r0 after store

	Memory Address	Memory Data
	0x20008007	0x00
	0x20008006	0x00
	0x20008005	0x00
	0x20008004	0x00
	0x20008003	0x00
	0x20008002	0x00
	0x20008001	0x00
r0 ⇨	0x20008000	0x00

Example ANS (Little-Endian ordering)

```
STR r1, [r0, #4]!
; r0 = 0x20008000,  r1=0x76543210
```

r0 before store

0x20008000

r0 after store

0x20008004

	Memory Address	Memory Data
	0x20008007	0x76
	0x20008006	0x54
	0x20008005	0x32
r0 ⇨	0x20008004	0x10
	0x20008003	0x00
	0x20008002	0x00
	0x20008001	0x00
	0x20008000	0x00

Addressing Modes for Load/Store Multiple Registers

Not Covered

```
STMxx rn{!}, {register_list}
LDMxx rn{!}, {register_list}
```

- xx = IA or DB (IB, DA: A32 only)

Addressing Modes	Description	Instructions
IA	Increment After	STMIA, LDMIA
IB (A32 only)	Increment Before	STMIB, LDMIB
DA (A32 only)	Decrement After	STMDA, LDMDA
DB	Decrement Before	STMDB, LDMDB

- **IA**: address is incremented by 4 after a word is loaded or stored.
- **IB**: address is incremented by 4 before a word is loaded or stored. (A32 only)
- **DA**: address is decremented by 4 after a word is loaded or stored. (A32 only)
- **DB**: address is decremented by 4 before a word is loaded or stored.

A32 only: IB and DA modes (STMIB/LDMIB, STMDA/LDMDA). Cortex-M (Thumb-2) supports only IA and DB.

Not Covered

Load/Store Multiple Registers

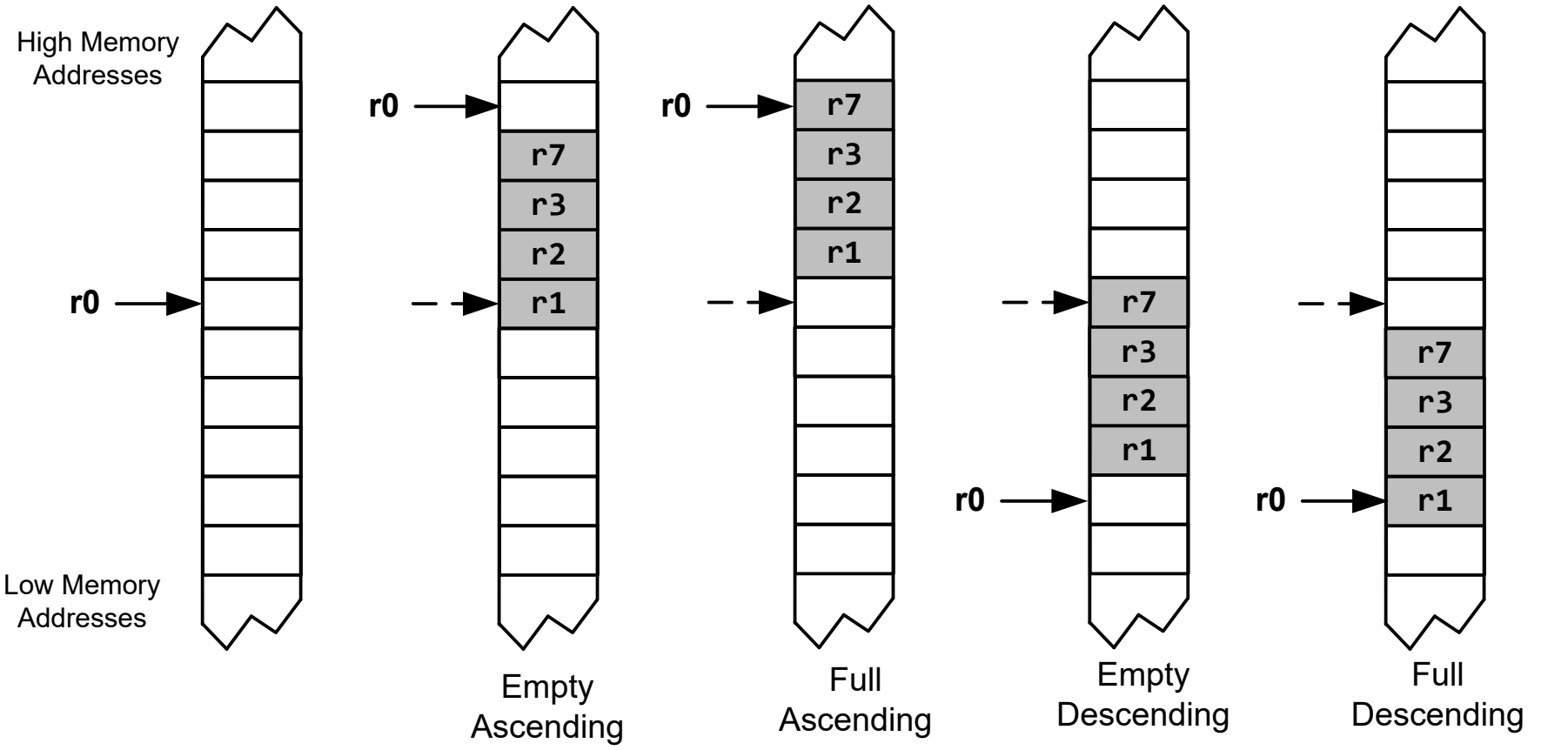
- The following are synonyms.
  - **`STM`** = **`STMIA`** (Increment After) = **`STMEA`** (Empty Ascending)
  - **`LDM`** = **`LDMIA`** (Increment After) = **`LDMFD`** (Full Descending)
  - Two independent switches:
    - Increment or decrement?
    - Before or after?
- The order in which registers are listed does not matter
  - For STM/LDM, the lowest-numbered register is stored/loaded at the lowest memory address.

A32 only: IB and DA modes (STMIB/LDMIB, STMDA/LDMDA). Cortex-M (Thumb-2) supports only IA and DB.

Not Covered

Store Multiple Registers

`STMxx r0!, {r3,r1,r7,r2}`

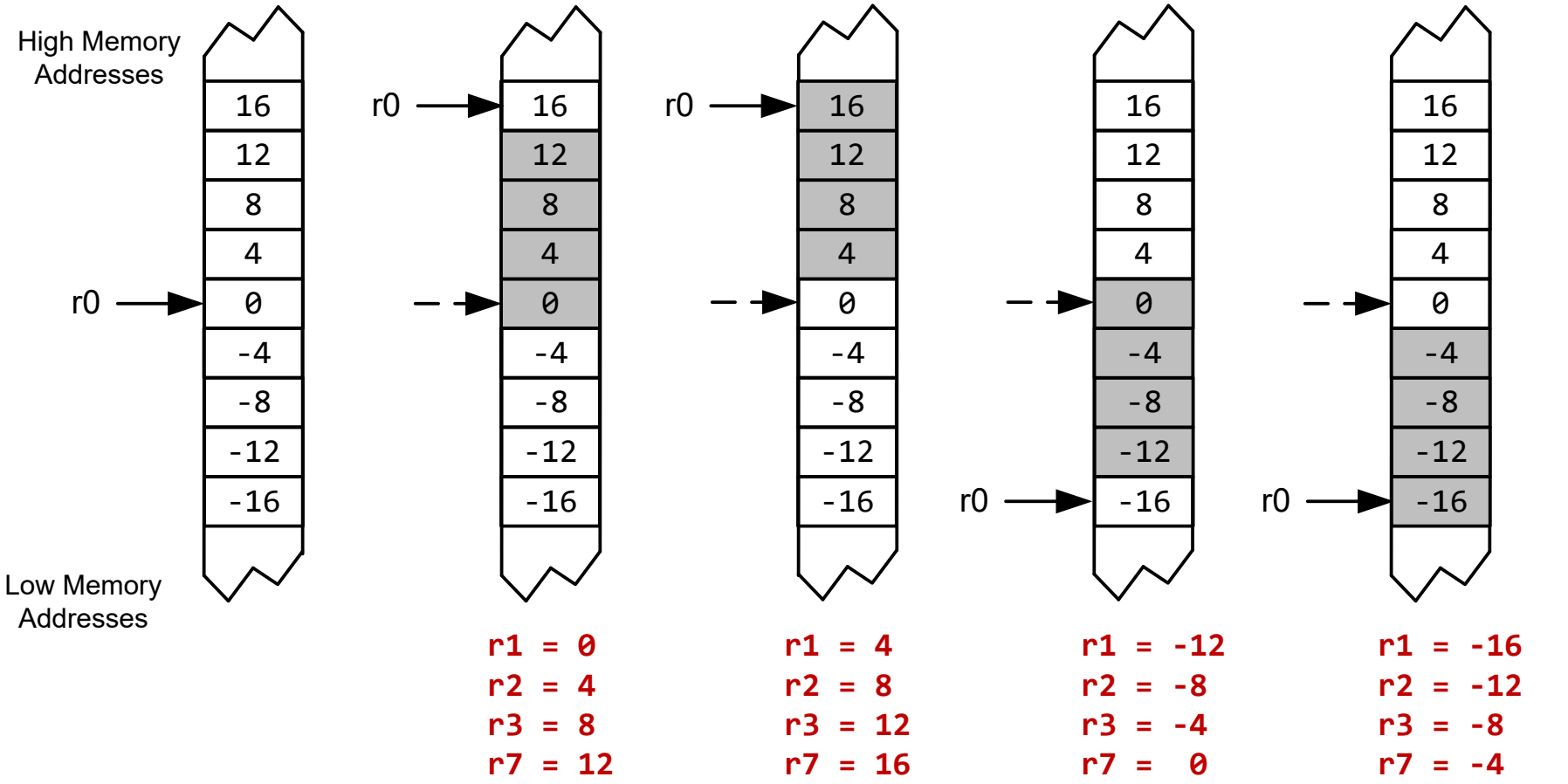


A32 only: IB and DA modes (STMIB/LDMIB, STMDA/LDMDA). Cortex-M (Thumb-2) supports only IA and DB.

Not Covered

Load Multiple Registers

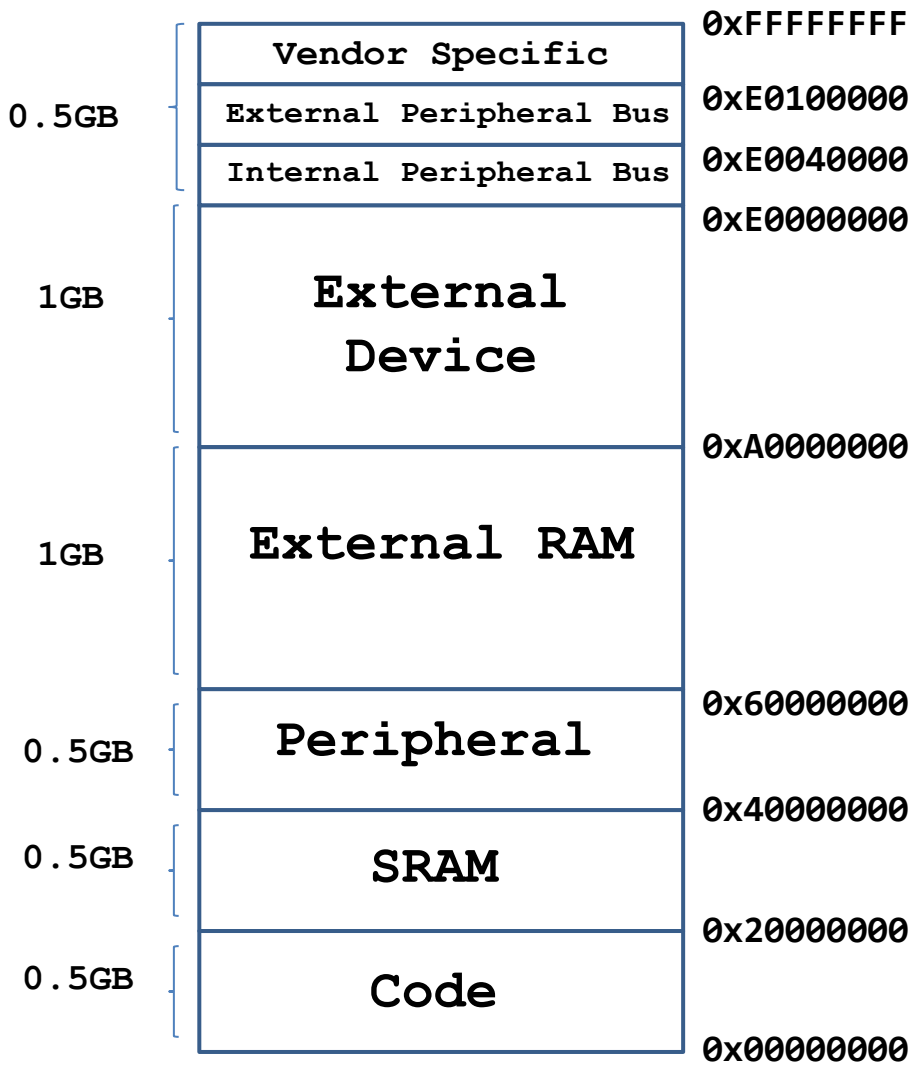
`LDMxx r0!, {r3,r1,r7,r2}`

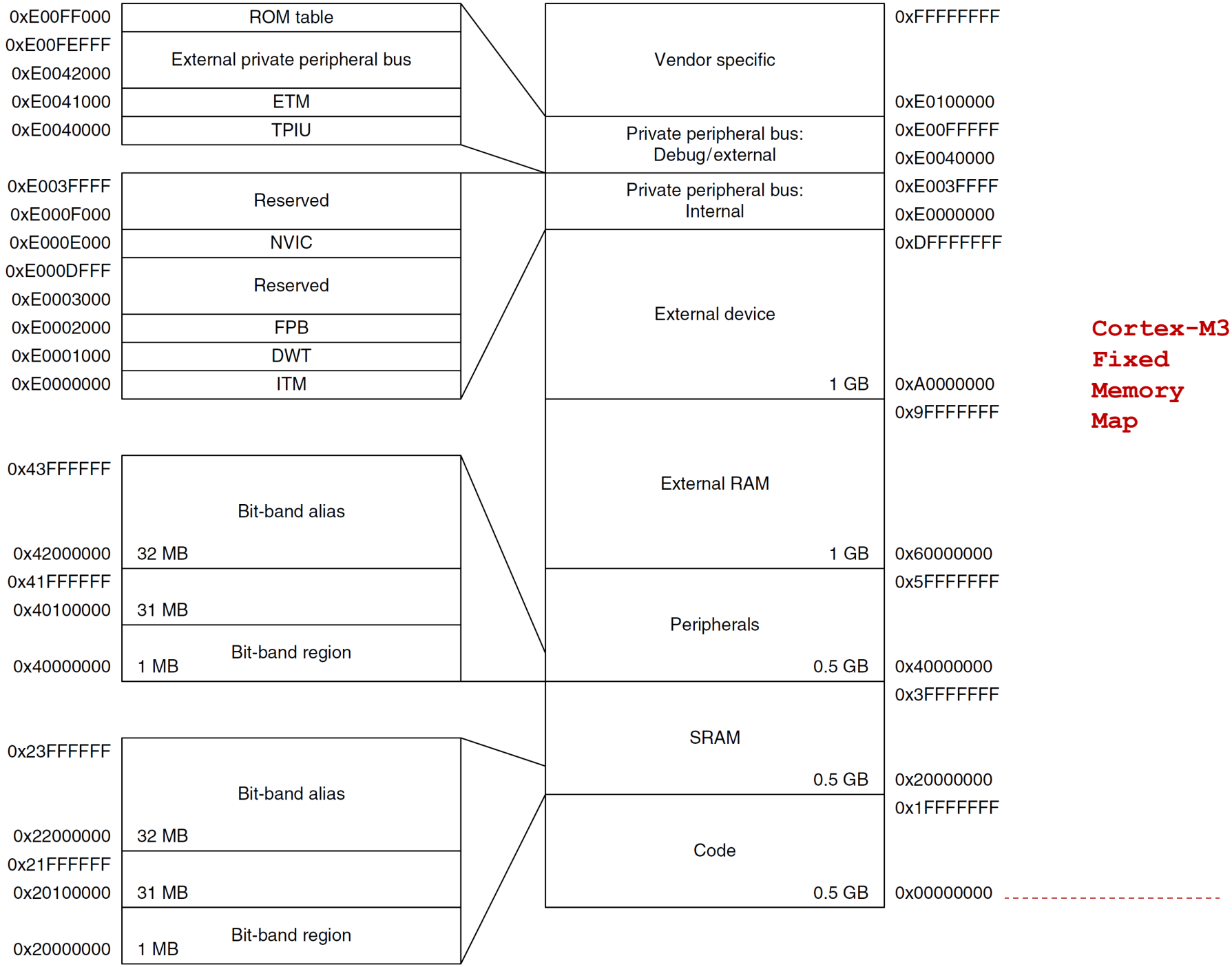


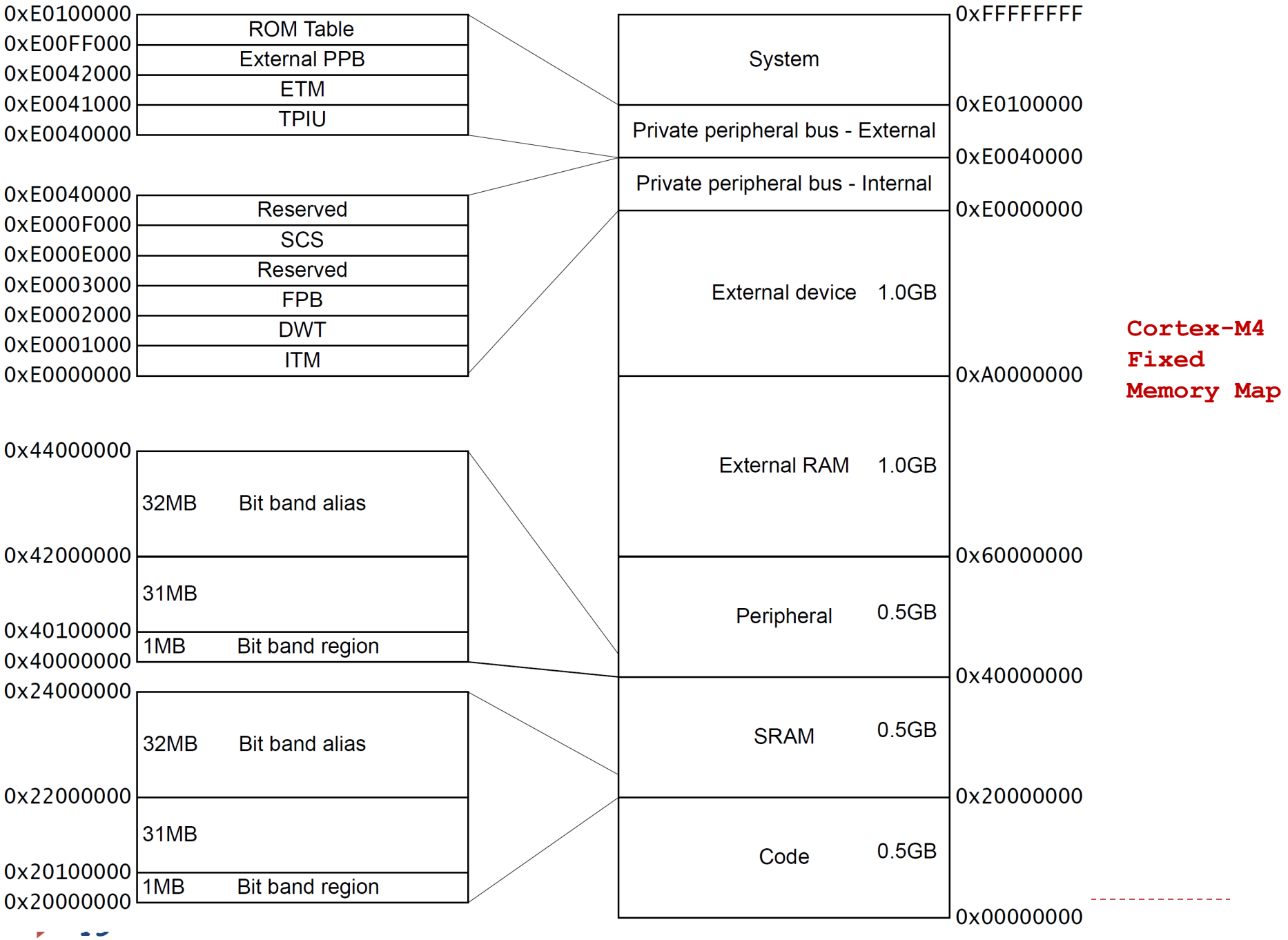
A32 only: IB and DA modes (STMIB/LDMIB, STMDA/LDMDA). Cortex-M (Thumb-2) supports only IA and DB.

Cortex-M3 & Cortex-M4 Memory Map

- 32-bit Memory Address
- 2^{32} bytes of memory space (4 GB)
- Harvard architecture: physically separated instruction memory and data memory
- Van Neumann architecture: unified instruction and data memory




0xE00FF000
ROM table
0xE00FEFFF
External private peripheral bus
0xE0042000
0xE0041000
ETM
0xE0040000
TPIU
0xE003FFFF
Reserved
0xE000F000
0xE000E000
NVIC
0xE000DFFF
Reserved
0xE0003000
0xE0002000
FPB
0xE0001000
DWT
0xE0000000
ITM
0x43FFFFFF
Bit-band alias
0x42000000
32 MB
0x41FFFFFF
0x40100000
31 MB
Bit-band region
0x40000000
1 MB
0x23FFFFFF
Bit-band alias
0x22000000
32 MB
0x21FFFFFF
0x20100000
31 MB
Bit-band region
0x20000000
1 MB
Vendor specific
0xFFFFFFFF
0xE0100000
Private peripheral bus:
Debug/external
0xE00FFFFF
0xE0040000
Private peripheral bus:
Internal
0xE003FFFF
0xE0000000
0xDFFFFFFF
External device
1 GB
0xA0000000
0x9FFFFFFF
External RAM
1 GB
0x60000000
0x5FFFFFFF
Peripherals
0.5 GB
0x40000000
0x3FFFFFFF
SRAM
0.5 GB
0x20000000
0x1FFFFFFF
Code
0.5 GB
0x00000000
Cortex-M3
Fixed
Memory
Map


0xE0100000
0xE00FF000
0xE0042000
0xE0041000
0xE0040000
ROM Table
External PPB
ETM
TPIU
0xE0040000
0xE000F000
0xE000E000
0xE0003000
0xE0002000
0xE0001000
0xE0000000
Reserved
SCS
Reserved
FPB
DWT
ITM
0x44000000
0x42000000
0x40100000
0x40000000
32MB Bit band alias
31MB
1MB Bit band region
0x24000000
0x22000000
0x20100000
0x20000000
32MB Bit band alias
31MB
1MB Bit band region
0xFFFFFFFF
0xE0100000
0xE0040000
0xE0000000
0xA0000000
0x60000000
0x40000000
0x20000000
0x00000000
System
Private peripheral bus - External
Private peripheral bus - Internal
External device 1.0GB
External RAM 1.0GB
Peripheral 0.5GB
SRAM 0.5GB
Code 0.5GB
Cortex-M4
Fixed
Memory Map

Pseudo-instructions

- Pseudo instruction: available to use in an assembly program, but not directly supported by hardware.
- Pseudo → not real
- Compilers translate it to one or multiple actual machine instructions
- Pseudo instructions are provided for the convenience of programmers.

LDR Pseudo-instruction

LDR Rt, =expr

LDR Rt, =label

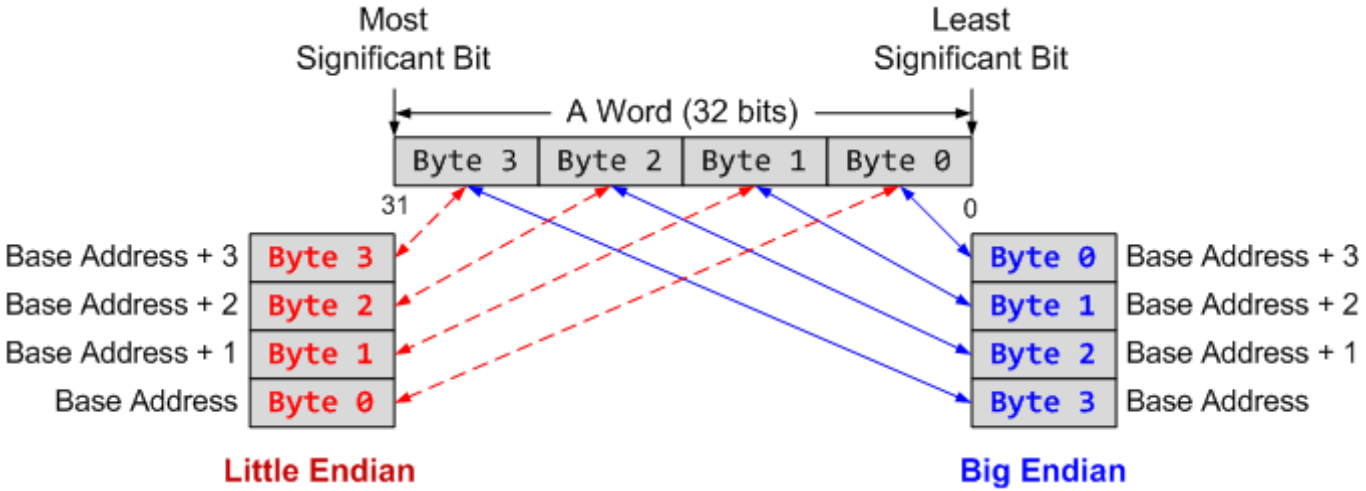
- If the value of expr can be loaded with **MOV**, **MVN** (16-bit instruction) or **MOVW** (32-bit instruction), the assembler uses that instruction.
  - MOV supports all 8-bit immediate numbers ranging in [0, 255]. For numbers out of this range, some patterns can be encoded.
- If a valid MOV, MVN, MOVW instruction cannot be used (due to out of range), or if the label_expr syntax is used, the assembler places the constant in a literal pool and generates a PC-relative LDR instruction that reads the constant from the literal pool.

```
LDR r1,=0xFF0   ; loads 0xFF0 into R1
                ; =>  MOV r1,#0xFF0
LDR r2,=0xFFF   ; loads 0xFFF into R2
                ; =>  MOVW r2, #0xFFF
LDR r3,=array   ; loads the address of array into R3
                ; =>  LDR r3,[pc, offset_to_litpool]
                ;     ...
                ;     litpool DCD array
```

Software uses this pseudo instruction to set a register to some value without worrying about the size of the value.

Summary

- Memory address is always in terms of bytes.
- How data is organized in memory?



- How data is addressed?

Addressing Format	Example	Equivalent
Pre-index	`LDR r1, [r0, #4]`	r1 ← memory[r0 + 4], r0 is unchanged
Pre-index with update	`LDR r1, [r0, #4]!`	r1 ← memory[r0 + 4] r0 ← r0 + 4
Post-Index	`LDR r1, [r0], #4`	r1 ← memory[r0] r0 ← r0 + 4

References

- Lecture 22. Big-Endian and Little-Endian
  - https://www.youtube.com/watch?v=T1C9Kj_78ek&list=PLRJhV4hUhIymmp5CCeIFPyxbknsdcXCc8&index=22
- Lecture 23. Load and Store Instructions
  - https://www.youtube.com/watch?v=CtfV3HsHwk4&list=PLRJhV4hUhIymmp5CCeIFPyxbknsdcXCc8&index=23
- Lecture 24. Addressing mode: pre-index, post-index, and pre-index with update
  - https://www.youtube.com/watch?v=zgkxPdPkxa8&list=PLRJhV4hUhIymmp5CCeIFPyxbknsdcXCc8&index=24
- ARM Instruction Set - Stack Instructions STMFD, STMFA , STMED, STMEA, Vishal Gaikwad (STMFA and STMED: A32 only)
  - https://www.youtube.com/watch?v=H4xoaOlNSJo